

Trainings ▾

Select Service Tools

Recommended Cummins® Service Tools

- Charge air Cooler Tester, Part Number 4919085, or equivalent
- Air Handling Clean Care Kit, Part Number 4919588, or equivalent
- Digital multimeter, Part Number 3164489, or equivalent
- Thermocouple adapter, Part Number 3164499, or equivalent
- Thermocouple Wire Kit, Part Number 3822988, or equivalent
- Pressure Gauge, Part Number 3823205

Additional Service Items

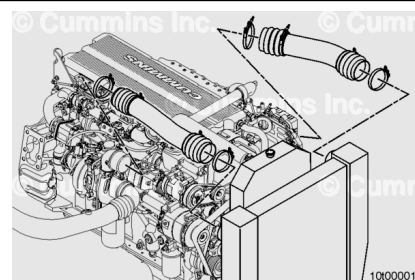
- Shop air
- Soapy water

Clean

If the engine experiences a turbocharger malfunction, or any other occasion where oil or debris is put into the charge air cooler, the charge air cooler **must** be cleaned.

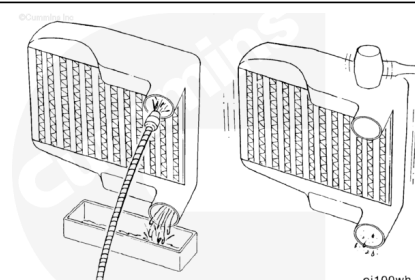
Remove the charge air cooler piping and charge air cooler from the vehicle. See equipment manufacturer service information.

Use protective caps from the Air Handling Clean Care Kit, Part Number 4919588, or equivalent, to cover open points on the plumbing and engine.



⚠ WARNING ⚠

When using solvents, acids, or alkaline materials for cleaning, follow the manufacturer's recommendations for use. Wear goggles and protective clothing to avoid personal injury.



⚠ CAUTION ⚠

**Do not use caustic cleaners to clean the charge air cooler.
Damage to the charge air cooler will result.**

Flush the charge air cooler internally with solvent in the opposite direction of normal airflow. Shake the charge air cooler and lightly tap on the end tanks with a rubber mallet to dislodge trapped debris. Continue flushing until all debris or oil is removed.

Note : Make sure the tubes are in the vertical direction when flushing.

If the debris can **not** be totally removed from the charge air cooler, the charge air cooler **must** be replaced.

⚠ WARNING ⚠

Wear appropriate eye and face protection when using compressed air. Flying debris and dirt can cause personal injury.

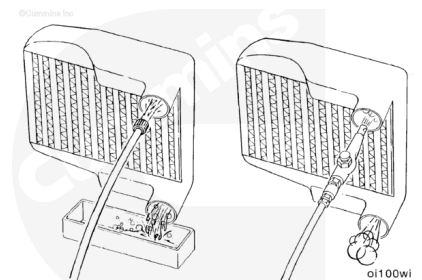
⚠ CAUTION ⚠

The charge air cooler must be rinsed, dried, and free of solvent, oil, and debris or engine damage will result.

After the charge air cooler has been thoroughly cleaned of all oil and debris with solvent, wash the charge air cooler internally with hot soapy water to remove the remaining solvent. Rinse thoroughly with clean water.

Blow compressed air through the inside of the charge air cooler in the opposite direction of normal air flow until the charge air cooler is dry internally.

Use protective caps from the Air Handling Clean Care Kit, Part Number 4919588, or equivalent, to cover open points on the charge air cooler after cleaning.

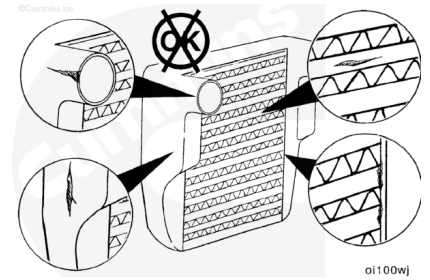


Inspect for Reuse

Inspect the charge air cooler for cracks, holes, or other damage.

Inspect the tubes, fins, and welds for tears, breaks, or other damage. If any damage causes the charge air cooler to **not** pass the air leak test, the charge air cooler **must** be replaced.

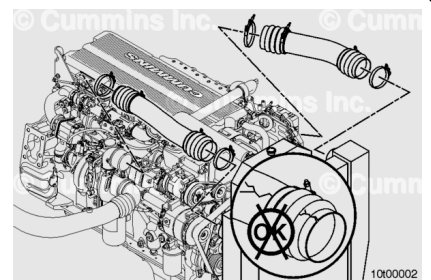
Install the charge air cooler on the vehicle. See equipment manufacturer service information.



Leak Test

⚠ CAUTION ⚠

All inlet and outlet hoses must be removed prior to pressurize the charge air cooler. Failing to do so may cause damage to the nature aspirated air compressor system.

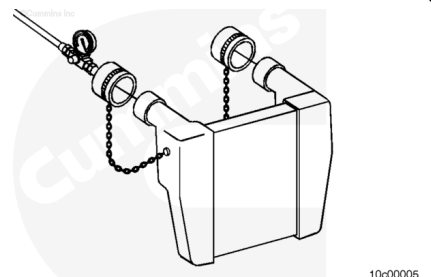


Note : The charge air cooler does **not** have to be removed from the chassis to complete this test.

To check the charge air cooler for cracked tubes or header, remove the inlet and outlet hoses from the charge air cooler.

⚠ WARNING ⚠

To prevent possible injury if either plug blows off during the test, secure safety chains on the test plugs to any convenient capscrew on the radiator assembly. This test must not be performed without securely fastened safety chains.

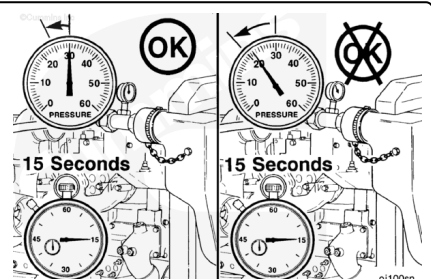


Install the charge air cooler tester, Part Number 4919085, or equivalent.

Apply 207 kPa [30 psi] of air pressure to the charge air cooler. Close the air pressure regulator.

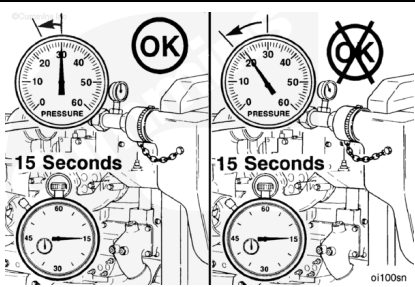
Determine if pressure decay is caused by a leak in the charge air cooler or from a leaky connection.

Use a spray bottle filled with soapy water and apply the solution to all hose connections. Watch for bubbles to appear, indicating the location of a leak.



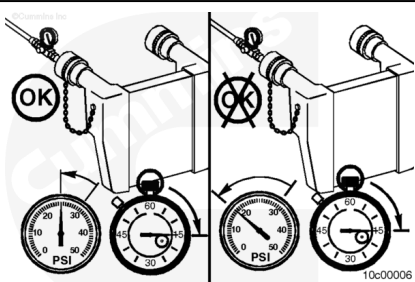
If the pressure decay is caused by a leaky connection, repair the connection and repeat the test.

If the leak is within the charge air cooler, repeat the test to verify the accuracy of the pressure decay measurement. Similar pressure decay readings **must** be obtained in at least three consecutive tests before the reading can be considered accurate.



Monitor the pressure gauge and determine the rate of pressure decay with a stopwatch.

Charge Air Cooler Pressure Drop		
kpa		psi
35	MAX	5



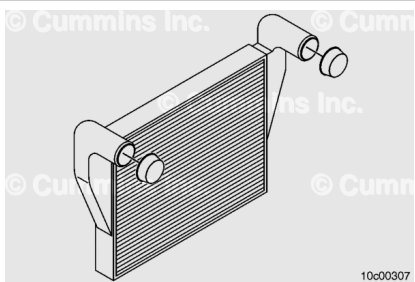
If the pressure drop is greater than the specification listed, the charge air cooler **must** be replaced.

See equipment manufacturer service information for replacement instructions.

Note : If a charge air cooler leaks more than 35 kPa [5 psi] in 15 seconds, it will appear as a major leak in a leak tank.

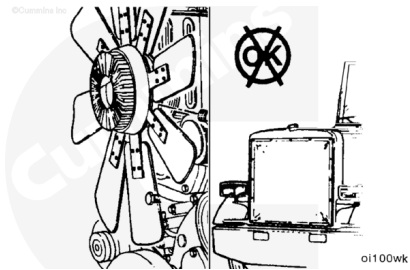
Note : Charge air coolers are **not** designed to be 100-percent leak-free. If the pressure decay is less than 35 kPa [5 psi] in 15 seconds, the charge air cooler does **not** need to be replaced.

After the test is complete, cover open points on the charge air cooler with protective caps from the Air Handling Clean Care Kit, Part Number 4919588, or equivalent.



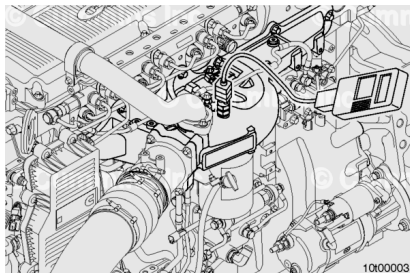
Temperature Differential Test

Inspect the charge air cooler fins for obstructions to airflow. Remove obstructions such as a winterfront or debris. Manually lock the shutters, if equipped, in the OPEN position.



Install an air probe thermocouple, Part Number 3164495, or equivalent, and connect to a Fluke™ digital multimeter, Part Number 3164489, or equivalent, using a temperature adapter, Part Number 3164499, or equivalent.

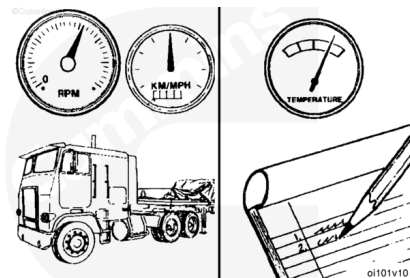
Install another thermocouple at the air cleaner inlet to the outside of the vehicle.



Perform a road test with the engine at peak power and a vehicle speed of 30 mph or greater.

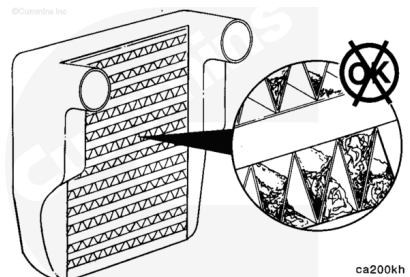
Record the intake air temperature and the ambient air temperature.

Calculate the differential temperature:



Intake Air Temperat ure	-	Ambient Air Temperat ure	=	Differenti al Temperat ure
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If the temperature differential is greater than the specifications, check the charge air cooler for dirt and debris on the fins and clean as necessary. If the problem still exists, check the charge air cooler for debris in the fins or between the charge air cooler and the radiator. Confirm full fan engagement.



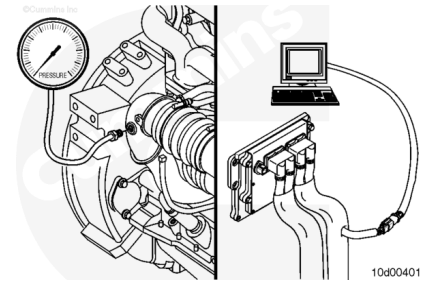
Charge Air Cooler Temperature Rise (ambient above
0°C [32°F])

celsius		fahrenheit
11	MAX	20

Pressure Differential Test

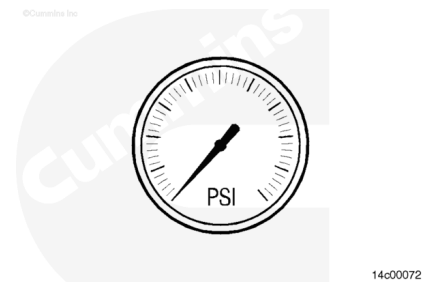
Use pressure gauge, Part Number 3823205, to record the air pressure at the inlet of the charge air cooler. Install the pressure gauge in the 1/8 - inch fitting in the turbocharger compressor outlet elbow.

Monitor the intake manifold pressure sensor reading with the electronic service tool, to record the air pressure at the outlet of the charge air cooler.



Operate the engine at the maximum advertised horsepower and correlating rpm. Record the readings on the pressure gauge.

Charge Air Cooler System Differential Pressure		
kpa		psi
21	MAX	3



If the differential pressure is greater than the specification, check the charge air cooler and associated piping for plugging. Clean or replace, if necessary.

Note : If a malfunction resulted in oil, excessive fuel, or excessive black smoke entering the exhaust system, the aftertreatment system must be inspected. Refer to the Aftertreatment Diesel Oxidation Catalyst and Aftertreatment Diesel Particulate Filter Reuse Guidelines, Bulletin 4021600 (</qs3/pubsys2/xml/en/bulletin/4021600.html>).

Note : If a malfunction resulted in coolant entering the exhaust system, the aftertreatment system can be recovered.

- Use the following procedure to check the aftertreatment diesel particulate filter (DPF). Refer to Procedure 014-016 in Section 14. (</qs3/pubsys2/xml/en/procedures/493/493-014-016.html>)

Note : If a malfunction resulted in coolant entering the exhaust system, the aftertreatment system can be recovered. Contact a Cummins® Authorized Repair Location.

